

Pragya Sharma

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EDUCATION

University of California Los Angeles, Los Angeles, CA

Sept 2021 - Mar 2027 (Exp)

Doctor of Philosophy in Electrical and Computer Engineering

Carnegie Mellon University, Mountain View, CA

Dec 2018

Master of Science in Electrical and Computer Engineering

VIT University, Vellore, India

May 2017

Bachelor of Technology in Electronics and Communication Engineering

WORK EXPERIENCE

NVIDIA Corporation, Santa Clara, CA

SYSTEMS SOFTWARE ENGINEERING INTERN, ML INFRA (ALPAMAYO)

Jun - Sept 2026

- Architected and implemented APOLLO end-to-end, an agentic system for investigating and validating performance optimizations in distributed, production-scale ML pipelines supporting AV foundation models.
- Engineered the investigation pipeline to reconstruct distributed executions, localize bottlenecks, test hypotheses in parallel across agents, and verify correctness and measurement validity before engineer review.
- Optimized production workloads for up to $8\times$ speedup and cut CPU consumption 40%-70% at output parity, saving a projected 71,000 CPU-hours weekly. Apollo is adopted across wider org for pipeline investigations.

University of California Los Angeles, Los Angeles, CA

GRADUATE RESEARCH ASSISTANT, NETWORKED & EMBEDDED SYSTEMS LAB

Sept 2021 - Present

Foundation-Model Serving and Orchestration

- Co-designed and built FMplex, a model virtualization layer giving each task a virtual foundation model with its own extensions, lifecycle, and isolation on a shared backbone; cut latency 80% versus spatial partitioning and hosted $6\times$ more tasks across 7 backbones (LLM, VLM, TSFM) and 92 tasks. ([ASPLOS '27, under review](#))
- Developed FMTK, a toolkit that composes time-series FM pipelines from reusable encoding, adaptation, and decoding components, reducing deployment code from 500 lines to 10. ([NeurIPS '25](#))
- Leading dissertation research on runtime orchestration of stateful, reasoning FM pipelines that shift placement across device, edge, and cloud as network and compute conditions change.

Task-Aware Distributed ML Systems

- Modeled distributed inference latency from sensing rate, platform throughput, network delay, and safety constraints; showed cloud inference holds application safety margins more reliably than on-device execution, overturning the assumption that latency-critical inference must stay local. ([ICCCN '25](#))
- Characterized device-edge-cloud placement of perception and control models in AV sense-to-act pipelines and designed Adaptive Control Placement, switching between local and cloud controllers on network conditions for 30% better time-to-collision. ([HotMobile '25, MILCOM '23](#))

Evaluation and Deployment Infrastructure

- Architected and open-sourced CADET, a distributed autonomy platform integrating CARLA, Mininet emulation, and heterogeneous compute (Jetson Orin, A6000) at <10 ms synchronization overhead with unified model, system, and task telemetry; proved V2V sharing eliminates collisions under tail delays. ([ICRA '26](#))
- Mentored the student leads on HoliBench and CREST, extensible frameworks where new devices, models, and inference backends plug into a fixed workflow for cross-platform FM characterization and HIL model design.

Amazon.com (Amazon Lab126), Sunnyvale, CA

EMBEDDED SOFTWARE ENGINEER

Feb 2019 - Aug 2021

- Developed protocol software and low-power device firmware for Amazon [Sidewalk](#), a 900 MHz IoT network now running across millions of Amazon, Ring, and third-party devices.
- Led design and implementation of the protocol configuration management feature, building an abstraction layer and APIs that moderate application-layer flash access and improve flash memory efficiency.

- Led low-latency firmware development for hardware-software integration, independently debugging critical issues and coordinating cross-functional teams to hit launch timelines.

Bosch LLC, R&D Department, Sunnyvale, CA

WIRELESS SYSTEMS INTERN

May 2018 - Aug 2018

- Engineered UWB node hardware, new localization and security algorithms for Perfectly Keyless, Bosch's passive entry system, reaching ~30 cm positioning accuracy for secure car lock/unlock.

Carnegie Mellon University, Mountain View, CA

GRADUATE RESEARCH ASSISTANT

Jan 2018 - Dec 2018

Automotive Wireless Security & V2V (Ford R&D, Argonne National Labs)

- Led a Ford-sponsored study demonstrating man-in-the-middle attacks on passive keyless entry systems at 300ft using software-defined radios, and engineered the supporting firmware and SDR hardware setup.
- Built an open-source DSRC/WAVE stack on BladeRF emulating vehicle-to-vehicle message exchange and proposed an algorithm to address the identified security vulnerabilities.

Session-Level Resource Allocation for Interactive Video

- Co-designed an admission control algorithm allocating resources to interactive video sessions while accommodating background traffic. As wireless lead built the 802.11g testbed that showed 47% higher throughput than conventional routers. ([IEEE JSAC '19](#))

SELECTED PUBLICATIONS

- *FMplex: Model Virtualization for Serving Extensible Foundation Models*
H. Shastri, **P. Sharma**, W. A. Hanafy, D. Irwin, M. B. Srivastava, P. Shenoy [[ASPLOS '27, under review](#)]
- *CADET: Enabling Cooperative Autonomy via Distributed Experimentation Toolkit*
P. Sharma, B. Wang, M. B. Srivastava [[ICRA '26](#)]
- *HoliBench: A Cross-Platform Benchmarking and Deployment Toolkit for FMs in CPS-IoT Applications*
I. Chakrabarti*, Z. Xiong*, **P. Sharma**, M. B. Srivastava [[arXiv '26](#)]
- *FMTK: A Modular Toolkit for Composable Time Series Foundation Model Pipelines*
H. H. Shastri, **P. Sharma**, P. Shenoy, M. B. Srivastava [[NeurIPS BERT^2S workshop '25](#)]
- *Cloud Is Closer Than It Appears: Revisiting the Tradeoffs of Distributed Real-Time Inference*
P. Sharma, H. Qiu, M. B. Srivastava [[ICCCN '25](#)]
- *Towards a Performance-Driven Device-Edge-Cloud Relationship*
P. Sharma, B. Wang, X. Ouyang, R. Nanayakkara, B. Balaji, P. Tabuada, M.B. Srivastava [[HotMobile 2025](#)]
- *Impact of Delays and Computation Placement on Sense-Act Application Performance in IoT*
P. Sharma, M. B. Srivastava [[MILCOM IoT workshop'23](#)]

LEADERSHIP EXPERIENCE

- Publication Chair, FMSys '26 (Workshop on Foundation Models for CPS & IoT, CPS-IoT Week)
- Mentored over 50 undergraduate and graduate students
- Mentor, The Women of Color Project
- Graduate Instructor, Los Angeles Computing Circle (CS Bootcamp 2022, 2023, 2024)
- Software Engineering Mentor, Amazon Mentorship Program
- Head, Women in ECE (WinECE), Carnegie Mellon University, Silicon Valley

TECHNICAL SKILLS

- **Programming Languages:** Python, Bash, C/C++, HTML/CSS/JavaScript, Assembly
- **AI/ML:** CUDA, PyTorch, TensorFlow, TensorRT, OpenCV, NCNN, LLMs, VLMs, Quantization
- **Infrastructure:** Spark, Slurm, Kubernetes, Docker, AWS, Mininet, CARLA, ROS, GCP, Maglev, Databricks
- **Hardware Platforms:** NVIDIA GPUs, Jetson, Raspberry Pi, AWS DeepRacer, Nordic, Semtech, TI
- **Wireless and Embedded:** LoRa, WiFi, Bluetooth/BLE, DSRC, Matter, FreeRTOS, UWB, Wireshark